

Electronic Devices and Circuits Lab

Assignment-3



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0.1 Introduction

This experiment aims to design and demonstrate the digital clock using Arduino for clock generation, seven-segment displays, D flip-flop counters, decoders and some logic gates to reset the counters. The whole project is operated by using the 1Hz frequency generated by Arduino to give it as the CLK to the initial counter and count the seconds, minutes and hours in this 8-hour digital clock.

The main objective is to demonstrate the functional digital clock that displays the seconds, minutes and hours.

0.2 Working Principle

0.2.1 Pulse Generation (Arduino UNO)

To give the clock to the circuit, we are using the pulse of 1Hz frequency generated by the Arduino UNO. This pulse is used in the MOD-10 counter of seconds to increment it. The Arduino Uno generates a 1 Hz pulse, which serves as the clock signal for the entire circuit. This pulse is used to increment the seconds counter in the clock.

0.2.2 Counting Mechanism

1. Seconds Counter :

The seconds counter (SN74LS74N D flip-flop IC) comprises a MOD 10 counter and a MOD 6 counter to count from 00 to 59 seconds of the clock. A 1Hz pulse is given by the Arduino and is fed into the first MOD-10 counter of the seconds counter, which then upon reaching the 9 sends out a carry-out pulse to the MOD-6 counter. Once the seconds counter reaches the 59 seconds mark, the counter resets the number to 00 and sends the carry-out pulse to the minutes counter.

2. Minutes Counter :

The carry-out pulse given by the Seconds counter is connected to the Minutes counter as a clock to increment the Minutes counter, which also comprises MOD-10 and MOD-6 counters using the SN74LS74N D flip-flops. This also works similarly to the seconds counter after reaching 59 from 00 it resets at 00 while sending the carry-out pulse to the hours counter.

3. Hours Counter :

The carry-out pulse given by the Minutes counter is connected to the Hours counter as a clock to increment the Hours counter, which comprises MOD-8 counters using the SN74LS74N D flip-flops. This counter also works similarly to the other two counters but this ranges only from 00 to 07. After reaching the seven it resets the counter to 00.

The BCD (Binary-Coded Decimal) output from each counter is given to the corresponding SN74LS47N BCD to the 7-segment decoder IC. This decoder is used to convert the BCD data into a form which is readable for the 7-segment displays, to display the

digits (00 to 07 for hours, and 00 to 59 minutes and seconds). The five 7-segment displays are connected to the output of these decoders where two displays represent the seconds, the other two minutes and the remaining one the hours.

0.2.3 Resetting using Gates

1. **7400 NAND Gate ICs** : These NAND gates are used to reset the hours, minutes and seconds without crossing the range of the desired values. Here we are supposed to give two inputs and the corresponding output is given to the $\overline{CLR}$ pins to reset the FlipFlop according to a certain logic.
2. **7404 NOT Gate ICs**: These NOT gates are also used to reset the hours without crossing the range of the desired values. For example, in the hours it should range from 0000 to 0111, so we need to reset the counter at 1000 we use a NOT gate to reset to 0000 when the MSB hits logic '1' by triggering the $\overline{CLR}$ pin.

0.3 Results

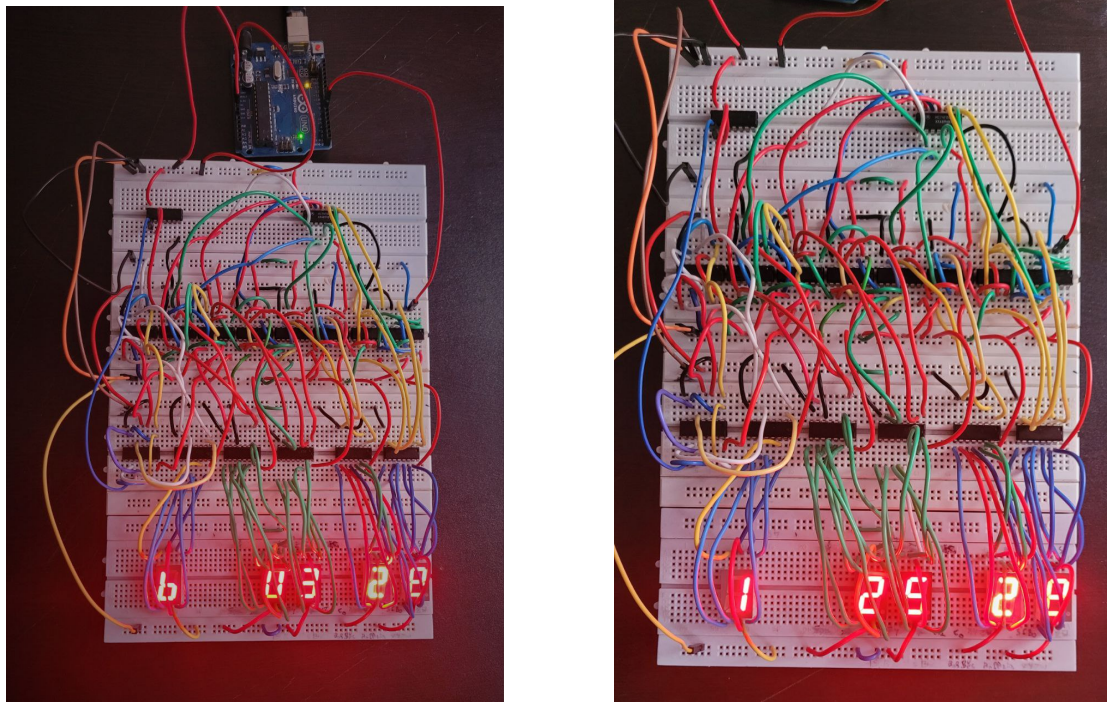


Figure 1: Experimental setup for the 8-hour digital clock project

0.4 Conclusion

In the above experiment, we have successfully demonstrated an 8-hour digital Clock using the Counters of D-FlipFlops, Arduino for 1Hz frequency, 7-segment decoders and displays where we achieved a working digital clock that accurately displays the time with the help of 7-segment displays with the help of pulse generated by Arduino. With

the help of this project, we understood the mechanism of counters and combinational logic circuits.

0.5 Precautions

- Ensure that all the counters are properly initialised.
- Ensure that all the wirings are made according to the schematic diagram, as miswiring may damage the component.